Building a global portfolio: what 18 years of evidence shows

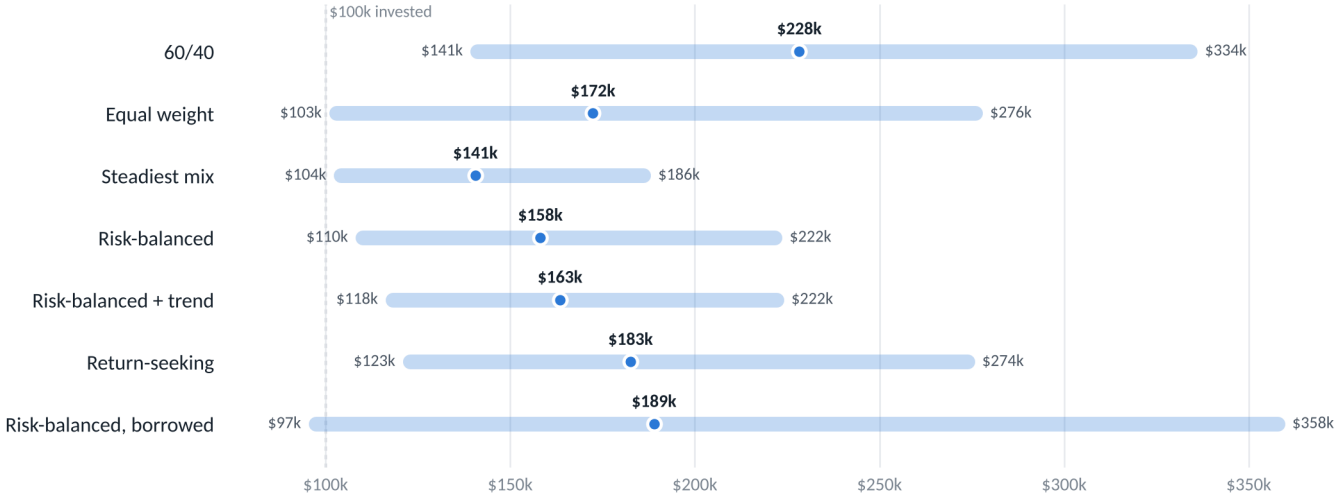
We tested six ways to combine ten global investments (U.S., developed and emerging-market stocks; Treasuries, corporate and inflation-protected bonds; commodities, gold and property) as if run month by month since 2008, using only what was known at the time and paying trading costs. We then asked whether market signals or a machine-learning model could improve on them, and simulated 10,000 possible decades.

WHAT WE FOUND

- **The plain 60/40 mix gave the most return for the risk taken**, earning 8.3% a year. The broader mixes earned 3.4% to 6.5%, because U.S. stocks, the 60/40's main holding, had an exceptional run.
- **Spreading money more widely cut the worst falls.** The risk-balanced mix fell at most 17% and the steadiest mix 14%, against 28% for the 60/40.
- **A simple trend rule helped a little.** Leaning toward investments that had beaten cash over the past year lifted the risk-balanced mix from 4.6% to 5.0% a year and cut its chance of a 20% fall over 10 years from 16% to 7%. The gain is small enough that it could be luck.
- **Our machine-learning model did not help.** On 164 months it never saw in training (2013 to 2026), it did worse than the simple trend rule and worse than making no changes at all. Its forecasts were no better than chance, so we would not use it.
- **Borrowing to boost the risk-balanced mix backfired.** It fell as much as 32% in 2008, and lost 22% in 2022 (the 60/40 lost 17%), when bonds and stocks fell together.

WHAT $100,000 COULD BECOME OVER 10 YEARS

Dot: middle outcome. Bar: 90 of 100 simulated decades fall inside it. Simulated by replaying blocks of the 2008–2026 record in random order.



TRACK RECORD, MAY 2008 TO SEPTEMBER 2026, AFTER TRADING COSTS

Approach	What it does	Return per year	Worst fall	Return per unit of risk	Chance of a 20%+ fall in 10 years
60/40	60% U.S. stocks, 40% U.S. Treasuries	8.3%	−28%	0.75	56%
Equal weight	10% in each of the ten investments	5.5%	−33%	0.45	48%
Steadiest mix	the mix with the smallest expected swings	3.4%	−14%	0.39	8%
Risk-balanced	each investment adds the same amount of risk	4.6%	−17%	0.49	16%
Risk-balanced + trend	risk-balanced, leaning toward what has been rising	5.0%	−16%	0.56	7%
Return-seeking	tilts toward recent winners, max 30% in any one	6.2%	−23%	0.59	38%
Risk-balanced, borrowed	risk-balanced, borrowing to match the 60/40's risk	6.5%	−32%	0.46	67%

What this means for you. Clever signals mattered far less than the basic mix. The right mix depends mostly on how large a temporary fall you can sit through without changing course. If a fall of 28% would not change your plans, the 60/40 rewarded patience. If it would, the risk-balanced mix with a trend tilt gave up about 3 points of return a year for a worst fall of 16%. The next decade may be less generous than the last: if every approach earns only half its past return above cash, the 60/40's middle outcome drops to 4.8% a year, with a 5% chance of ending the decade below where it started.

How it was tested. Each month-end, every approach set its mix from up to three years of daily prices (ten exchange-traded funds: SPY, EFA, EEM, IEF, LQD, HYG, TIP, DBC, GLD, VNQ; cash = BIL), then held it for a month. Trading costs of 0.02% to 0.10% per trade are deducted; doubling them changes no conclusion. The machine-learning model (gradient boosting) was retrained every January on past data only, with a gap between training and testing periods so it could not peek at the answers. *Return per unit of risk* is the Sharpe ratio. Data: Yahoo Finance, dividends reinvested, through October 9, 2026. **Important.** Past results and simulations do not predict future returns, and the period tested was unusually good for U.S. stocks. This note is for discussion and is not a recommendation for any individual.